```
adithya@LAPTOP-07A78KDI:~/OS/OS$ echo "Linux inode experiment" > original.txt
adithya@LAPTOP-07A78KDI:~/OS/OS$ ls -i original.txt
26572 original.txt
adithya@LAPTOP-07A78KDI:~/OS/OS$ stat original.txt
  File: original.txt
  size: 23              Blocks: 8          IO Block: 4096   regular file
Device: 8,48    Inode: 26572       Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/ adithya)   Gid: ( 1000/ adithya)
Access: 2026-10-16 05:04:15.015791786 +0000
Modify: 2026-10-16 05:04:15.015791786 +0000
Change: 2026-10-16 05:04:15.015791786 +0000
 Birth: 2026-10-16 05:04:15.015791786 +0000
adithya@LAPTOP-07A78KDI:~/OS/OS$ ln original.txt hardlink.txt
adithya@LAPTOP-07A78KDI:~/OS/OS$ ls -i original.txt hardlink.txt
26572 hardlink.txt  26572 original.txt
adithya@LAPTOP-07A78KDI:~/OS/OS$ ln -s original.txt symlink.txt
adithya@LAPTOP-07A78KDI:~/OS/OS$ ls -i original.txt hardlink.txt symlink.txt
26572 hardlink.txt  26572 original.txt  26573 symlink.txt
adithya@LAPTOP-07A78KDI:~/OS/OS$ ls -i original.txt hardlink.txt symlink.txt
26572 hardlink.txt  26572 original.txt  26573 symlink.txt
adithya@LAPTOP-07A78KDI:~/OS/OS$ find . -inum 123456
adithya@LAPTOP-07A78KDI:~/OS/OS$ find . -samefile original.txt
./original.txt
./hardlink.txt
adithya@LAPTOP-07A78KDI:~/OS/OS$
```

```c
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/mman.h>
#include <sys/stat.h>
#include <ctype.h>

int main(int argc, char *argv[]) {
    int fd;
    struct stat st;
    char *data;
    size_t i;

    if (argc != 2) {
        fprintf(stderr, "Usage: %s filename\n", argv[0]);
        return 1;
    }

    fd = open(argv[1], O_RDWR);
    if (fd == -1) {
        perror("open");
        return 1;
```

```
}

if (fstat(fd, &st) == -1) {
    perror("fstat");
    close(fd);
    return 1;
}

if (st.st_size == 0) {
    printf("File is empty.\n");
    close(fd);
    return 0;
}

data = mmap(NULL, st.st_size,
            PROT_READ | PROT_WRITE,
            MAP_SHARED, fd, 0);

if (data == MAP_FAILED) {
    perror("mmap");
    close(fd);
    return 1;
}

printf("Original content: ");
fwrite(data, 1, st.st_size, stdout);
printf("\n");

for (i = 0; i < (size_t)st.st_size; i++) {
    data[i] = (char)toupper((unsigned char)data[i]);
}

if (msync(data, st.st_size, MS_SYNC) == -1) {
    perror("msync");
    munmap(data, st.st_size);
    close(fd);
    return 1;
}

printf("Modified content: ");
fwrite(data, 1, st.st_size, stdout);
printf("\n");

if (munmap(data, st.st_size) == -1) {
```

```
        perror("munmap");
        close(fd);
        return 1;
    }

    close(fd);
    return 0;
}
```

```
adithya@LAPTOP-07A78KDI:~/OS/OS$ gcc -Wall -Wextra mmapfile.c -o mmap_file
adithya@LAPTOP-07A78KDI:~/OS/OS$ echo "Hello Linux Programming" > sample.txt
adithya@LAPTOP-07A78KDI:~/OS/OS$ ./mmap_file sample.txt
Original content: Hello Linux Programming

Modified content: HELLO LINUX PROGRAMMING

adithya@LAPTOP-07A78KDI:~/OS/OS$ cat sample.txt
HELLO LINUX PROGRAMMING
adithya@LAPTOP-07A78KDI:~/OS/OS$
```